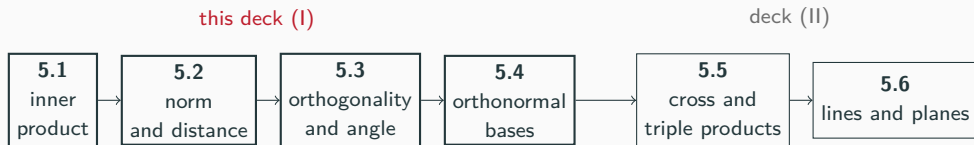


Linear Algebra

5 Inner product and geometry (I)

José Firmino Aguilar Madeira



Central idea: a vector space (Ch. 2) does not, by itself, have lengths or angles. The inner product adds them:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}, \quad \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Exercises: Sheet 5, Ex. 83–98 (global numbering continues from Sheet 4) and S1–S10 (geometry, deck II).

5.1 Inner product

Inner product and Euclidean space

Let U be a finite-dimensional real vector space.

Definition

An **inner product** on U is a map that assigns to each pair $(\mathbf{u}, \mathbf{v})$ a real number $\mathbf{u} \cdot \mathbf{v}$ such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in U$ and $\alpha, \beta \in \mathbb{R}$:

- i) $\mathbf{u} \cdot \mathbf{u} > 0$ if $\mathbf{u} \neq \mathbf{0}$ (positive definiteness)
- ii) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (symmetry)
- iii) $\mathbf{u} \cdot (\alpha \mathbf{v} + \beta \mathbf{w}) = \alpha (\mathbf{u} \cdot \mathbf{v}) + \beta (\mathbf{u} \cdot \mathbf{w})$ (linearity)

A finite-dimensional real vector space with an inner product is called a **Euclidean space**.

Other notation: $\langle \mathbf{u}, \mathbf{v} \rangle$. It is also called the 'scalar product' or 'dot product'.

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A finite-dimensional real vector space with an inner product is called a **Euclidean space**.

Other notation: $\langle \mathbf{u}, \mathbf{v} \rangle$. It is also called the 'scalar product' or 'dot product'.

Consequences: $\mathbf{0} \cdot \mathbf{v} = 0$; $(\alpha \mathbf{u} + \beta \mathbf{v}) \cdot \mathbf{w} = \alpha (\mathbf{u} \cdot \mathbf{w}) + \beta (\mathbf{v} \cdot \mathbf{w})$ (by ii and iii).

The standard inner product on $\mathbb{R}^n$

$$(x_1, \dots, x_n) \cdot (y_1, \dots, y_n) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

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Example. $(1, 2, -1) \cdot (3, 0, 2) = 3 + 0 - 2 = 1$.

It is not the only one. On $\mathbb{R}^2$, $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1 y_1 + x_2 y_2$ is also an inner product: it is symmetric, it is linear and $\langle \mathbf{x}, \mathbf{x} \rangle = 2x_1^2 + x_2^2 > 0$ if $\mathbf{x} \neq \mathbf{0}$.

In this course, unless stated otherwise, $\mathbb{R}^2$ and $\mathbb{R}^3$ carry the standard inner product.

- **Ex. 83** standard inner product on $\mathbb{R}^3$.
- **Ex. 84** with points: $\overrightarrow{AB} = B - A$; the condition gives an equation in a .

5.2 Norm and distance

Norm of a vector

Definition

The **norm** (length, magnitude) of $\mathbf{u}$ is $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$. On $\mathbb{R}^3$:

$$\|(x, y, z)\| = \sqrt{x^2 + y^2 + z^2}.$$

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Properties. For $\mathbf{u}, \mathbf{v} \in U$ and $\alpha \in \mathbb{R}$:

1. $\|\mathbf{u}\| \geq 0$, and $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$;

2. $\|\alpha\mathbf{u}\| = |\alpha| \|\mathbf{u}\|$;

3. $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$

(Cauchy–Schwarz inequality)

4. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

(triangle inequality)

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2. $\|\alpha\mathbf{u}\| = |\alpha| \|\mathbf{u}\|$;

3. $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ (Cauchy–Schwarz inequality)

4. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangle inequality)

A vector with norm 1 is called a **unit vector**. The **unit vector** in the direction of $\mathbf{u} \neq \mathbf{0}$ is $\text{vers } \mathbf{u} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$.

Example. $\|(1, 2, -2)\| = \sqrt{1 + 4 + 4} = 3$, $\text{vers}(1, 2, -2) = \left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right)$.

Definition

The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is the norm of their difference: $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

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Between two points A and B of $\mathbb{R}^3$: $d(A, B) = \|\overrightarrow{AB}\| = \|B - A\|$, that is,

$$d = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}.$$

Example. $A(1, 0, 2)$, $B(3, 1, 0)$: $\overrightarrow{AB} = (2, 1, -2)$, $d(A, B) = 3$.

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Example. $A(1, 0, 2)$, $B(3, 1, 0)$: $\overrightarrow{AB} = (2, 1, -2)$, $d(A, B) = 3$.

Properties.

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$, and $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$;
2. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$;
3. $d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w})$ (triangle inequality)

- **Ex. 85** distance between points: compare the squares ($d^2 = 17$).
- **Ex. 86** unit vector: $\|\mathbf{u}\|^2 = 1$. Read carefully what the equation says.

5.3 Orthogonality and angle

Orthogonal vectors

Definition

$\mathbf{u}$ and $\mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u} \cdot \mathbf{v} = 0$.

Properties (for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in U$, $\alpha \in \mathbb{R}$):

1. $\mathbf{0} \perp \mathbf{v}$;
2. $\mathbf{u} \perp \mathbf{v}$ and $\mathbf{u} \perp \mathbf{w} \Rightarrow \mathbf{u} \perp (\mathbf{v} + \mathbf{w})$;
3. $\mathbf{u} \perp \mathbf{v} \Rightarrow \mathbf{u} \perp \alpha \mathbf{v}$.

From 2 and 3: if $\mathbf{u}$ is orthogonal to $\mathbf{v}_1, \dots, \mathbf{v}_k$, it is orthogonal to every linear combination in $L\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

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Pythagoras' theorem

If $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. (proof: Ex. 91)

Angle between two vectors

Definition

For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the **angle** between $\mathbf{u}$ and $\mathbf{v}$ is the $\theta \in [0, \pi]$ such that

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

By Cauchy–Schwarz, the quotient lies in $[-1, 1]$, so θ exists and is unique.

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- $\mathbf{u} \cdot \mathbf{v} > 0$: acute angle; $= 0$: right angle; < 0 : obtuse.

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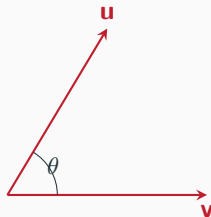
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- $\mathbf{u} \cdot \mathbf{v} > 0$: acute angle; $= 0$: right angle; < 0 : obtuse.

Example. $\mathbf{u} = (1, 1, 0)$, $\mathbf{v} = (1, 0, 1)$:

$$\cos \theta = \frac{1}{\sqrt{2} \sqrt{2}} = \frac{1}{2}, \quad \theta = \frac{\pi}{3} \text{ (60°)}.$$



- **Ex. 87** checking orthogonality; in b), use the note on linear combinations.
- **Ex. 88** general formulas for the cosine and the distance in $\mathbb{R}^3$, and a numerical case.
- **Ex. 91** Pythagoras: expand $\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v})$.
- **Ex. 97** angles of a triangle: at each vertex, both vectors *start* at that vertex.
- **Ex. 98** given angle: after squaring, check the sign of $\mathbf{v} \cdot \overrightarrow{AB}$.

5.4 Orthonormal bases

Definition

A basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ of a Euclidean space is **orthonormal** (o.n.) if its vectors are unit vectors and pairwise orthogonal:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

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Advantage: the coordinates are obtained from inner products, without solving systems (Ch. 2):

$$\mathbf{u} = (\mathbf{u} \cdot \mathbf{e}_1) \mathbf{e}_1 + \dots + (\mathbf{u} \cdot \mathbf{e}_n) \mathbf{e}_n.$$

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Example. $\mathbf{e}_1 = \frac{1}{5}(3, 4)$, $\mathbf{e}_2 = \frac{1}{5}(-4, 3)$ is an o.n. basis of $\mathbb{R}^2$. For $\mathbf{u} = (5, 10)$:

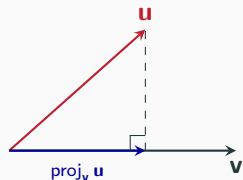
$$\mathbf{u} \cdot \mathbf{e}_1 = \frac{15+40}{5} = 11, \quad \mathbf{u} \cdot \mathbf{e}_2 = \frac{-20+30}{5} = 2, \quad \mathbf{u} = 11 \mathbf{e}_1 + 2 \mathbf{e}_2.$$

Orthogonal projection and Gram–Schmidt

Projection of $\mathbf{u}$ onto $\mathbf{v} \neq \mathbf{0}$:

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}, \quad \mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u} \perp \mathbf{v}.$$

Gram–Schmidt idea: from each basis vector, subtract its projections onto the previous ones; at the end, divide each one by its norm.



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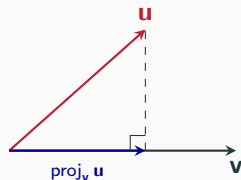
Gram–Schmidt idea: from each basis vector, subtract its projections onto the previous ones; at the end, divide each one by its norm.

Example in $\mathbb{R}^2$: basis $\{(3, 1), (2, 2)\}$.

$$\mathbf{v}_1 = (3, 1), \quad \mathbf{v}_2 = (2, 2) - \frac{(2, 2) \cdot (3, 1)}{(3, 1) \cdot (3, 1)}(3, 1) = (2, 2) - \frac{8}{10}(3, 1) = \left(-\frac{2}{5}, \frac{6}{5}\right).$$

O.n. basis: $\left\{ \frac{1}{\sqrt{10}}(3, 1), \frac{1}{\sqrt{10}}(-1, 3) \right\}.$

With three vectors it works the same way: this is what Ex. 95 and 96 ask for.



More generally: projection onto a subspace

Extension (not required in any exercise of Sheet 5): it brings orthonormal bases, coordinates and projection together in a single idea.

Orthogonal projection onto V

If $(\mathbf{e}_1, \dots, \mathbf{e}_k)$ is an orthonormal basis of a subspace V :

$$\text{proj}_V \mathbf{u} = (\mathbf{u} \cdot \mathbf{e}_1) \mathbf{e}_1 + \dots + (\mathbf{u} \cdot \mathbf{e}_k) \mathbf{e}_k, \quad \mathbf{u} - \text{proj}_V \mathbf{u} \perp V.$$

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Example. $V = L\{(1, 1, 0), (0, 0, 1)\}$, o.n. basis $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $\mathbf{e}_2 = (0, 0, 1)$; $\mathbf{u} = (3, 1, 2)$:

$$\text{proj}_V \mathbf{u} = \frac{4}{\sqrt{2}} \mathbf{e}_1 + 2 \mathbf{e}_2 = (2, 2, 0) + (0, 0, 2) = (2, 2, 2), \quad \mathbf{u} - \text{proj}_V \mathbf{u} = (1, -1, 0) \perp \mathbf{e}_1, \mathbf{e}_2.$$

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Example. $V = L\{(1, 1, 0), (0, 0, 1)\}$, o.n. basis $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $\mathbf{e}_2 = (0, 0, 1)$; $\mathbf{u} = (3, 1, 2)$:

$$\text{proj}_V \mathbf{u} = \frac{4}{\sqrt{2}} \mathbf{e}_1 + 2 \mathbf{e}_2 = (2, 2, 0) + (0, 0, 2) = (2, 2, 2), \quad \mathbf{u} - \text{proj}_V \mathbf{u} = (1, -1, 0) \perp \mathbf{e}_1, \mathbf{e}_2.$$

$\text{proj}_V \mathbf{u}$ is the vector of V closest to $\mathbf{u}$. This is the idea behind the least-squares method (fitting a line to data).

- **Ex. 89, 90** standard basis; computations with norm, angle and distance.
- **Ex. 92** equidistant vectors, given angles and an o.n. basis of $\mathbb{R}^3$.
- **Ex. 93, 94** Gram–Schmidt with two vectors: first in general, then in $\mathbb{R}^2$.
- **Ex. 95, 96** the same with three vectors: in general, then in $\mathbb{R}^3$.

LINEAR ALGEBRA IN PRACTICE: cosine similarity

A text can be represented by the vector of counts of a few words. With the words 'matrix', 'vector', 'football':

$$\mathbf{d}_1 = (3, 2, 0), \quad \mathbf{d}_2 = (6, 4, 0), \quad \mathbf{d}_3 = (0, 1, 4).$$

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The cosine of the angle measures similarity, *regardless of the length* of the text:

$$\cos(\mathbf{d}_1, \mathbf{d}_2) = 1 \quad (\text{same topic, text twice as long}), \quad \cos(\mathbf{d}_1, \mathbf{d}_3) = \frac{2}{\sqrt{13}\sqrt{17}} \approx 0.13.$$

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- This is how search engines and recommender systems compare documents, with thousands of coordinates.
- In physics, the **work** done by a force $\mathbf{F}$ along a displacement $\mathbf{d}$ is
$$W = \mathbf{F} \cdot \mathbf{d} = \|\mathbf{F}\| \|\mathbf{d}\| \cos \theta.$$

LINEAR ALGEBRA ON THE COMPUTER: inner product and QR

For illustration only — not examinable. Python (NumPy); in MATLAB/Octave: `dot`, `norm`, `[Q,R] = qr(A)`.

```
import numpy as np
u = np.array([1, 2, -1])
v = np.array([3, 0, 2])
u @ v                # 1
np.linalg.norm([1,2,-2]) # 3.0

# columns (3,1) and (2,2):
A = np.array([[3, 2],
              [1, 2]])
Q, R = np.linalg.qr(A)
Q.T @ Q              # = I
```

$$Q \approx \begin{bmatrix} -0.949 & -0.316 \\ -0.316 & 0.949 \end{bmatrix}$$

The columns of Q are $-\frac{1}{\sqrt{10}}(3, 1)$ and $\frac{1}{\sqrt{10}}(-1, 3)$: the o.n. basis we obtained by hand (5.4), up to sign.

$A = QR$, with R upper triangular: this is the Gram–Schmidt process written with matrices (**QR factorisation**).

Concept	How to obtain it / what it says
Inner product	positive definite, symmetric, linear; on $\mathbb{R}^n$: $\sum x_i y_i$
Norm	$\ \mathbf{u}\ = \sqrt{\mathbf{u} \cdot \mathbf{u}}$; $\ \alpha \mathbf{u}\ = \alpha \ \mathbf{u}\ $
Unit vector (vers)	$\mathbf{u}/\ \mathbf{u}\ $ (unit vector with the same direction and orientation as $\mathbf{u}$)
Distance	$d(\mathbf{u}, \mathbf{v}) = \ \mathbf{u} - \mathbf{v}\ $; $d(A, B) = \ B - A\ $
Inequalities	$ \mathbf{u} \cdot \mathbf{v} \leq \ \mathbf{u}\ \ \mathbf{v}\ $ (Cauchy–Schwarz); $\ \mathbf{u} + \mathbf{v}\ \leq \ \mathbf{u}\ + \ \mathbf{v}\ $
Orthogonal	$\mathbf{u} \cdot \mathbf{v} = 0$; Pythagoras: $\ \mathbf{u} + \mathbf{v}\ ^2 = \ \mathbf{u}\ ^2 + \ \mathbf{v}\ ^2$
Angle	$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\ \mathbf{u}\ \ \mathbf{v}\ }$, $\theta \in [0, \pi]$
O.n. basis	$\mathbf{e}_i \cdot \mathbf{e}_j = 0$ ($i \neq j$), $= 1$ ($i = j$); coordinates $\mathbf{u} \cdot \mathbf{e}_i$
Projection	$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$
Gram–Schmidt	subtract the projection onto the previous vectors; normalise
Proj. onto V	$\sum_i (\mathbf{u} \cdot \mathbf{e}_i) \mathbf{e}_i$, ($\mathbf{e}_i$) o.n. basis of V

KEY TAKEAWAYS — Chapter 5

1. The inner product is what gives a vector space lengths, distances and angles.
2. Norm: $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$; distance: $\|\mathbf{u} - \mathbf{v}\|$; angle: via the cosine.
3. Two vectors are orthogonal when their inner product is zero; then Pythagoras' theorem holds.
4. In an orthonormal basis, the coordinates of a vector are inner products.
5. In a finite-dimensional Euclidean space, any basis can be turned into an orthonormal basis by the Gram–Schmidt process.

Next (deck II): cross and scalar triple products, lines and planes in $\mathbb{R}^3$.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many worked exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 5 (exercises 83–98 and S1–S10) and its solutions.